

InverCRS: Generative Audio Inversion Attack in Collaborative Recognition Systems

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Abstract—Audio recognition systems have become integral to various applications, including speech-to-text, virtual assistants, and security monitoring, where efficiency and privacy are key concerns. The collaborative recognition system (CRS) partitions and deploys the neural network (NN) across multiple edge devices for cooperative recognition without sharing raw audio data. This distributed approach significantly alleviates the computational burden on the client and ensures data privacy. These advantages make CRS increasingly prevalent in audio recognition applications to improve security, efficiency, and provide timely feedback. However, sharing information during collaboration still poses the risk of exposing original data. To the best of our knowledge, this paper introduces *InverCRS*, the first inversion attack targeting CRS-empowered audio recognition systems. *InverCRS* is a generative attack in which the attacker trains a local generative model to take intermediate results as input and output the original audio. Once the generative model is trained, it can perform audio inversion using new intermediate results without the need for further optimization. Furthermore, *InverCRS* utilizes heuristic algorithms to approximate gradients, making it applicable to both white-box and black-box scenarios. We conduct comprehensive experiments to evaluate the feasibility and efficiency of *InverCRS* across two real-world audio datasets. The results demonstrate that *InverCRS* can effectively reconstruct the original audio from various split points within the CRS. Additionally, we investigate two potential defense strategies and provide experimental evaluations of their effectiveness in mitigating this attack.

Index Terms—collaborative inference, audio inversion, generative model, white-box and black-box attack

I. INTRODUCTION

Audio recognition has become a cornerstone of modern artificial intelligence, driving applications such as voice assistants [20], automated transcription services [3], and biometric authentication [29]. These systems leverage neural networks to process audio data, which is characterized by its temporal dynamics and spectral richness. Such features, while enabling accurate recognition, also demand substantial computational resources, especially for real-time audio processing.

To alleviate the computational burden on resource-constrained devices, many audio recognition systems rely on cloud-based neural networks [2], [5], [12], where users offload processing tasks to powerful remote servers. However, this

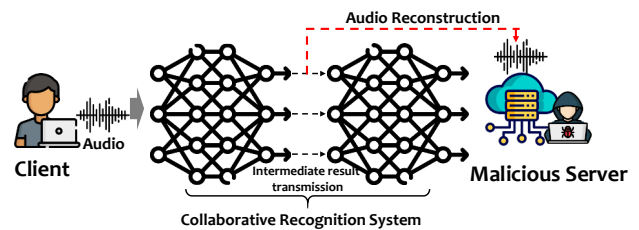


Fig. 1: Illustration of the inversion attack in the collaborative audio recognition system. The malicious server can reconstruct the client's original voice signal by receiving the intermediate calculation results sent by the client to achieve privacy theft.

approach necessitates the transmission of raw audio data to the cloud, raising significant privacy concerns [1]. Collaborative recognition systems (CRS) [13], [15] have emerged as a promising solution, dividing the neural network between the client and server to strike a balance between efficiency and privacy. By processing raw data locally through the initial layers and transmitting only intermediate representations to the cloud, CRS reduces client-side computational demands while mitigating the risk of directly exposing raw data.

Despite the advantages of CRS, recent research has revealed a critical vulnerability: intermediate representations can be exploited to reconstruct the original input, posing a significant threat to data privacy [18], [44], [46]. While inversion attacks have been extensively investigated in the domain of computer vision where attackers can reconstruct high-quality images from intermediate representations similar vulnerabilities in audio recognition systems remain comparatively underexplored. The temporal and spectral complexity of audio data poses challenges for inversion, yet successful attacks could reveal sensitive information like spoken content or speaker identity. This gap in research underscores the need for a deeper understanding of generative inversion attacks in audio recognition systems within the CRS paradigm.

In this paper, we introduce *InverCRS*, the first generative audio inversion attack targeting collaborative CRS. The goal of *InverCRS* is to reconstruct audio signals from the intermediate representations generated during CRS processes. Figure 1 illustrates the attack scenario of *InverCRS*, where the attacker

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(i.e., malicious server) exploits intermediate results received from the victim (i.e., client) in CRS to reconstruct the original audio data. Unlike traditional inversion attacks in computer vision, which rely on iterative optimization (e.g., gradient descent) for each new intermediate result, InverCRS eliminates this computationally expensive step. Prior methods [18], [46] face significant challenges in large-scale CRS applications, as they require re-optimization for every intermediate result, incurring substantial computational overhead. InverCRS addresses these limitations by enabling attackers to locally train a generative model. Once trained, the generative model can directly reconstruct original audio from new intermediate results without iterative optimization.

InverCRS is designed to be effective in both white-box and black-box scenarios. In the white-box scenario, the attacker acts as a malicious server, collecting intermediate results from the victim to train a generative attack model. Crucially, this process does not require access to the original audio corresponding to these intermediate results. Once the model is trained, it can directly reconstruct the original audio from new intermediate inputs. In the black-box scenario, where the attacker lacks access to the target model's structure and parameters, only the latter half of the split model is accessible. Traditional optimization techniques, such as gradient descent, are inapplicable in this setting, which needs accessing the whole model. To overcome this, a heuristic optimization strategy is employed to approximate gradients, enabling the effective training of the attack model. Notably, the training process for InverCRS does not rely on access to the target model's training dataset or any dataset with a similar distribution (i.e., auxiliary dataset). This independence makes it highly practical for real-world scenarios, where such data is often restricted due to privacy concerns or logistical challenges. In summary, the contributions of our work are as follows:

- We investigate the feasibility of reconstructing original audio from intermediate results transmitted in CRS and introduce InverCRS, the first generative audio inversion attack designed specifically for CRS.
- InverCRS is effective in both white-box and black-box settings without requiring any auxiliary datasets. In the white-box scenario, attackers leverage the target model's architecture and parameters to train the attack model using gradient descent. In the black-box scenario, InverCRS uses heuristic algorithms to approximate gradients and update the attack model iteratively.
- Comprehensive experiments on two real-world datasets validate the effectiveness of InverCRS. These experiments also examine the impact of the split point depth in the CRS on the InverCRS's performance. Furthermore, we explore several potential defense mechanisms and conduct experimental evaluations to demonstrate the robustness of InverCRS against these defenses.

II. RELATED WORK

In this section, we briefly introduce related work, including model inversion attacks and attacks against audio models.

A. Model Inversion Attack

Model inversion attack focuses on the privacy theft of training set attributes and reconstructs the input data by the model output. In model inversion attack [14], the attacker optimizes the quality of the inversion data through the confidence of the output and reduces the gap between the inversion data and the target input in the output confidence. In work [39], the attacker designed the embedding-reverse Generative adversarial network (GAN) technique, which can recover the facial features of the victim through the face recognition system. Khosravy et al. [22] proposed an efficient model inversion attack by integrating a pre-trained deep face generator, which efficiently mapped the model inversion attack search space into face feature vectors instead of general image space. Yang et al. [43] proposed a novel model inversion in an adversarial setting via background knowledge alignment. [32] analyzed the "MI overfitting" phenomenon, which prevents reconstructed images from learning the semantics of the training data, and proposed a method of model augmentation to enhance the inversion attack. [31] proposed a model inversion attack that only requires predicted labels, based on knowledge transfer. Han et al. [16] proposed a reinforcement learning-based black-box model inversion attack, which formulates the latent space search as a Markov Decision Process (MDP) problem and solves it with reinforcement learning. Zhou et al. [47] enhanced model inversion attacks by injecting adversarial examples into the training process of the attack to increase sample diversity.

With the popularization of smart devices and the expansion of NN structure, distributed machine learning has attracted more and more attention, e.g., collaborative inference [13], [40] and federated learning [25]. However, under these frameworks, the generated intermediate parameters become a breeding ground for model inversion attacks. Hitaj et al. [19] proposed that attackers can maliciously upload error gradients in collaborative learning to contaminate the global model, which could induce the target participant to release more class information while training a GAN model locally to extract information. Work [41] shows that in federated learning, the server can isolate the victim, and further recover the training set based on the gradient uploaded by the victim. [18], [44], [46] reveals that in the collaborative inference system, the attacker can recover the original data through the intermediate value. However, most of the existing work has only focused on inversion attacks in computer vision.

B. Attacks on Audio Models

Next, we will introduce attacks on audio models, including adversarial examples and audio inversion attacks.

Kreuk et al. [23] first introduced adversarial perturbations into the field of speaker verification. Liu et al. [28] focused on the number and weight of distortions to enhance the attack. Li et al. [24] proposed a universal adversarial perturbation generative network for speaker recognition, which maps the normal distribution into the subspace of universal adversarial perturbations to ensure the perturbations follow a standard normal distribution. Chen et al. [8] introduced PhoneyTalker,

an out-of-the-box toolkit for generating universal and transferable adversarial example attacks on speaker recognition systems. Xie et al. [42] designed FAPG, which uses Wave-U-Net to generate adversarial perturbations for audio inputs in a single forward pass, improving efficiency. Unlike previous approaches, Qu et al. [36] developed audio adversarial examples entirely from scratch, rather than adding perturbations.

Additionally, several studies have focused on black-box audio adversarial perturbations. Chen et al. [7] employed a gradient descent-based algorithm to solve the optimization problem in black-box settings, leveraging a natural evolution strategy to approximate the gradient. Chen et al. [10] proposed Devil's Whisper, a method for generating physical adversarial attacks against commercial black-box speech recognition devices, ensuring the generated audio adversarial examples are practical in real-world conditions. Yuan et al. [45] introduced CommanderSong, which stealthily embeds voice commands into songs to control ASR systems. CommanderSong has the potential to spread through the Internet and radio, influencing millions of ASR users. Chen et al. [9] proposed Metamorph, which injects inaudible commands into over-the-air voice to disrupt ASR systems. They first collect a small amount of distorted audio to generate an initial audio perturbation. Li et al. [26] considered the effects of speaker and microphone limitations, absorption and reverberation, and environmental noise in the adversarial perturbation generation process to enhance robustness. Qin et al. [35] proposed an imperceptible, robust, and targeted adversarial example for automatic speech recognition, utilizing the psychoacoustic principle [27] of auditory masking to add adversarial perturbations in regions inaudible to humans. The work by Schönherr et al. [38] used room impulse responses (RIRs) to generate robust audio adversarial examples tailored to arbitrary room characteristics. These audio adversarial examples are robust in over-the-air scenarios and independent of the source.

Moliner et al. [30] presents CQT-Diff, a data-driven generative audio model that can, once trained, be used for solving various different audio inverse problems in a problem-agnostic setting. Work [33] proposes an inversion attack that reverses the original audio through gradients, which can be effectively extended to distributed learning frameworks such as federated learning. Pizzi et al. [34] proposed the sliding model inversion, which is the extension of the standard inversion attack. The attack could reconstruct audio samples from a trained ML model and extract intermediate voice feature representations. Overall, most of the existing work focuses on adversarial example attacks against audio models, with little attention paid to audio inversion attacks.

III. PRELIMINARY

In this section, we introduce collaborative inference systems and methods for converting vectors into audio.

A. Collaborative Recognition System

In the collaborative recognition system (CRS), the NN model is divided into multiple parts and allocated to various

devices to execute the complete recognition process. Without loss of generality, in this paper, we only consider the scenario where the model f_θ is divided into two parts: $f_\theta = f_{\theta_1} \circ f_{\theta_2}$. The part from the input layer to the split point (i.e., f_{θ_1}) is placed on the client side, while the part from the split point to the output layer (i.e., f_{θ_2}) is placed on the server side. In CRS, the placement of the split point often needs to consider the following factors:

- **Computational Overhead:** The primary objective of CRS is to offload most of the computational overhead in the NN inference process to the server side. However, when the split point is placed deeper within the network, the client side bears a greater portion of the computational load. As a result, in CRS, the split point is typically positioned in a shallower layer to minimize client-side overhead. Nevertheless, due to privacy concerns, it is not feasible to fully delegate the NN's recognition tasks to the server side because, in this case, clients should upload raw data for recognition.
- **Latency:** To achieve faster recognition speed, the volume of data transmitted from the client side to the server side should be minimized as much as possible. The primary factor influencing the transmission volume is the dimensionality of the intermediate result vector transmitted at the split point. Therefore, to reduce latency, the split point should be strategically placed in an intermediate layer of the NN that has a smaller output dimension.
- **Storage Overhead:** When performing recognition tasks, the model must be loaded into the device's memory. Since clients often have limited storage capacity, such as mobile phones and smart wearable devices, the placement of split points in CRS must be carefully considered to minimize storage overhead on the client side.

Considering the factors outlined above, the placement of split points in CRS can be viewed as an optimization problem. In most cases, the majority of the layers of the neural network are allocated to the cloud server.

B. Methods for Converting Vectors into Audio

In model inversion attacks, attackers often need to convert the vectors obtained through inversion back into original input signals, such as images, audio, and so on. There are two methods to convert vectors into audio signals, as follows:

- **If the vectors are MFCCs:** First, the Mel-frequency cepstral coefficients (MFCCs) must be converted back into a magnitude spectrogram. This is accomplished through the inverse Mel-frequency cepstrum (iMFC) transformation, which applies the inverse discrete cosine transform (iDCT) to each frame of the MFCCs. This step reconstructs the Mel-scaled spectrogram and typically requires knowledge of the number of coefficients retained during the MFCC extraction process to ensure accuracy. Next, the Mel-scaled spectrogram must be transformed back into a time-domain signal. This is done by performing the inverse short-time Fourier transform (iSTFT). In this process, the phase information, if

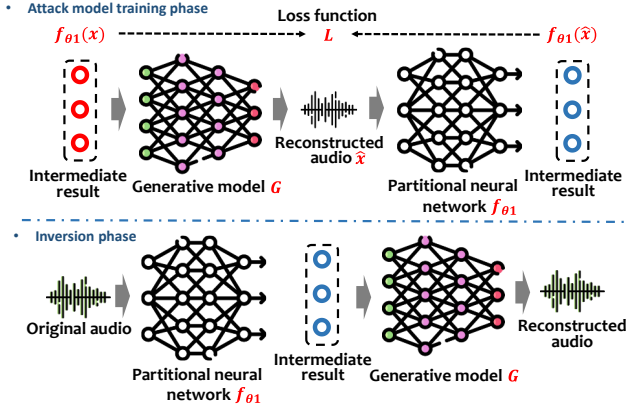


Fig. 2: Workflow of InverCRS in the audio CRS. InverCRS is mainly divided into two phases: the attack model training phase and the inversion phase. In the attack model training phase, the attacker collects a large number of intermediate results to train the generative attack model G . In the reversion phase, the attacker receives the victim's intermediate result and inputs it to G to reverse the original audio.

available from the original signal, is incorporated. In cases where the original phase information is unavailable, phase approximation techniques such as the Griffin-Lim algorithm or iterative phase reconstruction methods are employed to estimate the missing phase data. Finally, the output of the iSTFT produces the reconstructed audio signal in the time domain, thereby completing the inversion process. This approach ensures a robust reconstruction of the audio signal while addressing the challenges associated with phase recovery and spectral resolution.

- **If the vectors are Mel-spectrograms:** The Mel-spectrogram is typically derived from the Short-Time Fourier Transform (STFT) of the audio signal. To reconstruct the time-domain audio signal from the mel-spectrogram, the inverse Short-Time Fourier Transform (iSTFT) is used. The iSTFT processes the spectrogram, incorporating phase information, to synthesize the corresponding audio waveform in the time domain.

IV. INVERCRS IMPLEMENTATION

This chapter outlines the threat model and the implementation of InverCRS in both white-box and black-box scenarios. As illustrated in Figure 2, the InverCRS framework is divided into two distinct phases: the attack model training phase and the inversion phase. Both phases are included in the workflow for both white-box and black-box scenarios, with each tailored to the respective setting.

A. Threat Model

This section introduces the threat model of InverCRS, specifically including the adversary's goal, and the adversary's knowledge in both white-box and black-box scenarios.

Adversary's Goal: Define the target audio recognition model as f_θ , where θ is the model parameter. In CRS, f_θ is

divided into two parts: $f_\theta = f_{\theta_1} \circ f_{\theta_2}$. The client holds f_{θ_1} , and the server holds f_{θ_2} . During the process of collaborative audio recognition, f_{θ_1} in the client receives audio x and computes the intermediate result $f_{\theta_1}(x)$, then the client sends intermediate results $f_{\theta_1}(x)$ to the server. The malicious server would train a generative attack model locally and reconstruct the client's original audio based on the intermediate results.

Adversary's Knowledge: The knowledge available to the attacker, i.e., malicious server, varies in white-box and black-box scenarios. (I) In the white-box scenario, we assume that the attacker has access to both the model's parameters and structure, enabling the attacker to fully obtain f_{θ_1} . For instance, edge device providers may deploy part of the model on edge devices and the other part on the company's cloud server, where the cloud server can access the complete model information. (II) In the black-box scenario, we assume that the attacker can only query the output of the model f_{θ_1} and does not have access to its internal parameters or structure. For example, the model f_θ owner divides and encapsulates the model to f_{θ_1} and f_{θ_2} , assigning different portions to the client side and the malicious server to provide API services. In the black-box scenario, the malicious server may register an auxiliary user account, similar to other legitimate clients, with the ability to make queries but without access to the model f_{θ_1} 's internal parameters or structure. We further assume that the attacker cannot access the training set of the target model or any dataset with the same distribution as the training set to train a substitute model. This is a reasonable assumption, as the training set and datasets with the same distribution may contain private information, such as voiceprints, messages, or other sensitive data.

B. White-box Attack

In the white-box scenario, the attacker can access the structure and parameters of the model f_{θ_1} . In the attack model training phase, the attacker first collects a large number of intermediate results $\{f_{\theta_1}(x_i)\}_{i=1}^n$, where x_i is the original audio corresponding to the intermediate result. These intermediate results contain a large amount of semantic characteristics of the original audio. It should be emphasized that the attacker cannot directly obtain the collected original audio dataset $\{x_i\}_{i=1}^n$ due to privacy protection. For ease of description, we define $f_{\theta_1}(x_i)$ as I_i next.

After collecting the intermediate results, the attacker initializes the attack model G locally, which is a generative model. Its input size is the same as the size of the intermediate result, and its output size is the same as the corresponding size of the original audio. In this paper, we use the transposed convolutional neural network as the attack model G to reconstruct the original audio. The loss function for training the generative attack model G is as follows:

$$L = \frac{1}{n} \sum_{i=1}^n \|G(I_i) - x_i\|_2. \quad (1)$$

However, the optimization process of Equation (1) requires the attacker to access the original audio dataset $\{x_i\}_{i=1}^n$, which

Algorithm 1 The attack model training in InverCRS to attack collaborative recognition systems in white-box scenario.

```

1: function MODELTRAIN( $f_{\theta_A}, \{x_i\}_{i=N}, lr, \epsilon, T, k$ )
2:   /*  $f_{\theta_A}$  is the victim's model */
3:   /*  $\{x_i\}_{i=N}$  is the set of audio of the victim */
4:   /*  $\lambda$  is the equilibrium coefficient in E.q. (4) */
5:   /*  $G$  is the generative attack model */
6:   /*  $lr$  is the learning rate */
7:   /*  $T$  is the number of iterations */
8:   /*  $k$  is BatchSize */
9:   /*  $\theta_G$  represent the parameter of  $G$  */
10:
11: Client Side(Victim):
12:   Compute intermediate result  $\{f_{\theta_1}(x_i)\}_{i=N}$ ;
13:   Transmit  $\{f_{\theta_1}(x_i)\}_{i=N}$  to server;
14: Server Side(Attacker):
15:   initialize  $G, t \leftarrow 0$ 
16:   while  $t < T$  do
17:     Randomly draw  $k$  samples  $\{f_{\theta_1}(x_j)\}_{j=k}$  from the set
        $\{f_{\theta_1}(x_i)\}_{i=N}$ .
18:     Calculate the loss function  $L$  according to E.q. (4).
19:     Calculate the gradient  $g_{\theta_G}$  according to E.q. (5).
20:     Update parameters  $\theta_G$ :  $\theta_G = \theta_G - lr * g_{\theta_G}$ .
21:      $t = t + 1$ 
22:   end while
23: return generative attack model  $G$ 
24: end function

```

is impractical in reality due to privacy-preserving. To address this issue, we have modified the loss function L as follows:

$$L = \frac{1}{n} \sum_{i=1}^n \|f_{\theta_1}(G(I_i)) - I_i\|_2. \quad (2)$$

Specifically, a high-quality reconstructed audio $\hat{x}_i = G(I_i)$ must have similar characteristics to the original audio x_i . Therefore, by minimizing the difference between the corresponding intermediate results of $\hat{x}_i$ and x_i , the generative attack model G can also be effectively trained.

To minimize the difference between the reconstructed audio and the original, we introduce Total Variation Denoising (TVD) to reduce noise in the reconstructed audio. The goal of TVD is to eliminate high-frequency noise components by minimizing the total variation of the audio signal, while preserving the key features of the original audio. Abrupt changes in the audio signal often correspond to important information, so the TVD method is designed to protect these critical frequency features, thereby making the reconstructed audio sound more natural. The TVD (denoted as $V(s)$ with the audio s) is formulated as follows:

$$V(s) = \sum_{j=1}^{r-1} |s_{i+1} - s_i|, \quad (3)$$

where $s_{i+1} - s_i$ represents the change between the i -th and $(i+1)$ -th sampling points, and r is the total number of sampling points in the audio. After introducing the Total Variation Denoising, the loss function is as follows:

$$L = \frac{1}{n} \sum_{i=1}^n \|f_{\theta_1}(G(I_i)) - I_i\|_2 + \lambda \times \frac{1}{n} \sum_{i=1}^n V(G(I_i)), \quad (4)$$

where the equilibrium coefficient λ is to balance these two terms. If the split point is deeper, meaning that f_{θ_1} contains

more neural network layers, a larger λ is used. This is because the intermediate results output by deeper layers will lose more original audio information, and directly reconstructing the audio using Equation (2) will contain more noise. Therefore, more noise needs to be denoised. Then, the generative attack model G can be trained by the gradient descent algorithm. The corresponding gradient g_{θ_G} calculation for its parameters θ_G is as follows:

$$g_{\theta_G} = \frac{\partial L}{\partial \theta_G} = \frac{1}{n} \sum_{i=1}^n \frac{\partial \|f_{\theta_1}(G(I_i)) - I_i\|_2}{\partial \theta_G} + \lambda \times \frac{1}{n} \sum_{i=1}^n \frac{\partial V(G(I_i))}{\partial \theta_G}. \quad (5)$$

Algorithm 1 sketches the complete process of training the generative attack model in the white-box scenario. Once the attack model G is trained, when the attacker receives the new intermediate result $f_{\theta_1}(x_t)$, he can directly input $f_{\theta_1}(x_t)$ into G to reconstruct the corresponding audio $\hat{x}_t = G(f_{\theta_1}(x_t))$. Finally, the attacker adopts the methods of Section III-B to convert the vector to audio.

C. Black-box Attack

In the black-box scenario, we assume that the attacker cannot access the parameters and structure of model f_{θ_1} , nor can they obtain the training dataset of the model f_{θ} or a dataset that has identically distributed. Meanwhile, we assume that the attacker could masquerade as another legitimate client query the output of model f_{θ_1} , that is, the attacker could input audio into f_{θ_1} to obtain the corresponding intermediate result and utilize the services of CRS. In addition, the attacker is able to collect intermediate results transmitted from the victim client side, but cannot access the corresponding original audio. From Equation (5), we can derive:

$$\frac{\partial \|f_{\theta_1}(G(I_i)) - I_i\|_2}{\partial \theta_G} = \frac{\partial \|f_{\theta_1}(G(I_i)) - I_i\|_2}{\partial G(I_i)} \times \frac{\partial G(I_i)}{\partial \theta_G}. \quad (6)$$

Obviously, in the black-box scenario, the attacker cannot utilize Equation (5) to update the parameters of the attack model because the attacker can not compute $\frac{\partial \|f_{\theta_1}(G(I_i)) - I_i\|_2}{\partial G(I_i)}$ without accessing the parameter θ_1 of model f_{θ_1} .

To tackle this problem, we leverage the heuristic algorithm, Natural Evolution Strategies (NES), to estimate the gradient $\frac{\partial \|f_{\theta_1}(G(I_i)) - I_i\|_2}{\partial G(I_i)}$. NES can optimize the objective function without any knowledge about the specific parameters of the objective function. Given this property, NES has been widely used in optimization problems [4], [6], [21]. NES estimates the gradient by maximizing the expected value of the objective function under the Gaussian distribution. Next, we present how to use NSE to estimate the gradient $\frac{\partial \|f_{\theta_1}(G(I_i)) - I_i\|_2}{\partial G(I_i)}$.

Firstly, the attacker randomly selects $2m$ Gaussian noises $\delta_1, \delta_2, \dots, \delta_{2m}$ as the search direction by symmetric sampling, which satisfy $\delta_i = -\delta_{2m+1-i}$ for $i \in \{1, 2, \dots, 2m\}$. Subsequently, these noises are multiplied by the search variance σ and added to the reconstructed audio $\hat{x}_j = G(I_j)$ to generate $2m$ new audio signals $x_{j,1}, x_{j,2}, \dots, x_{j,2m}$, where

Algorithm 2 The attack model training in InverCRS to attack collaborative recognition systems in black-box scenario.

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1: function MODELTRAIN( $f_{\theta_A}, \{x_i\}_{i=N}, lr, \epsilon, T, k$ )
2:   /*  $f_{\theta_A}$  is the victim's model */
3:   /*  $\{x_i\}_{i=N}$  is the set of audio of the victim */
4:   /*  $\lambda$  is the equilibrium coefficient in E.q. (4) */
5:   /*  $G$  is the generative attack model */
6:   /*  $lr$  is the learning rate */
7:   /*  $T$  is the number of iterations */
8:   /*  $k$  is BatchSize */
9:   /*  $\theta_G$  represent the parameter of  $G$  */
10:
11: Client Side(Victim):
12:   Compute intermediate result  $\{f_{\theta_1}(x_i)\}_{i=N}$ ;
13:   Transmit  $\{f_{\theta_1}(x_i)\}_{i=N}$  to server;
14: Server Side(Attacker):
15:   initialize  $G, t \leftarrow 0$ 
16:   while  $t < T$  do
17:     Randomly draw  $k$  samples  $\{f_{\theta_1}(x_j)\}_{j=k}$  from the set
        $\{f_{\theta_1}(x_i)\}_{i=N}$ .
18:     Calculate the loss function  $L$  according to E.q. (4).
19:     Symmetrically sample  $2m$  Gaussian noises.
20:     for  $f_{\theta_1}(x_j) \in \{f_{\theta_1}(x_j)\}_{j=k}$  do
21:       Approximate  $\frac{\partial \|f_{\theta_1}(G(I_j)) - I_j\|_2}{\partial G(I_j)}$  using E.q. (7).
22:       Compute  $\frac{\partial G(I_j)}{\partial \theta_G}$  and  $\frac{\partial V(G(I_j))}{\partial \theta_G}$ .
23:     end for
24:     Calculate the gradient  $g_{\theta_G}$  according to E.q. (5).
25:     Update parameters  $\theta_G$ :  $\theta_G = \theta_G - lr * g_{\theta_G}$ .
26:      $t = t + 1$ 
27:   end while
28: return generative attack model  $G$ 
29: end function

```

$\hat{x}_{j,i} = \hat{x}_j + \sigma \delta_i$. Then the derivative $\frac{\partial \|f_{\theta_1}(G(I_i)) - I_i\|_2}{\partial G(I_i)}$ could be approximated via NES as follows:

$$\frac{\partial \|f_{\theta_1}(G(I_i)) - I_i\|_2}{\partial G(I_i)} \approx \frac{1}{2m \times \sigma} \sum_{j=1}^{2m} \delta_j \times \|f_{\theta_1}(G(I_i) + \sigma \delta_j) - I_i\|_2. \quad (7)$$

Note that $f_{\theta_1}(G(I_i) + \sigma \delta_j)$ is the output of f_{θ_1} that can be queried by the attacker, and I_i is the collected intermediate result. We could substitute Equation (7) for $\frac{\partial \|f_{\theta_1}(G(I_i)) - I_i\|_2}{\partial G(I_i)}$ in Equation (6). Since $\frac{\partial G(I_i)}{\partial \theta_G}$ and $\frac{\partial V(G(I_i))}{\partial \theta_G}$ are not involved in the model f_{θ_1} , the attacker can directly compute and obtain them. In conclusion, the gradient g_{θ_G} can be approximated through NES. Algorithm 2 presents the training process of the generative attack model in the black-box scenario.

V. EXPERIMENTAL EVALUATION

This section presents the datasets and models required for the experiments, metrics, and the performance of InverCRS in both white-box and black-box scenarios.

A. Dataset and model

We conducted experimental evaluations of InverCRS on two real-world audio datasets: UrbanSound8K [37] and Google speech commands [11]. For each data, we first transform the audio into Mel-spectrogram or MFCC vectors and then make

Dataset	Target Model	Accuracy	Split Point
UrbanSound8K	6 conv + 2 fc	77.26%(Mel)	- 1st activation layer (ReLU1)
		64.82%(MFCC)	- 2nd activation layer (ReLU2)
Google speech commands	Resnet18	95.65%(Mel) 94.20%(MFCC)	- 4th activation layer (ReLU4)
			- 6th activation layer (ReLU6)
			- 1st maxpool layer
			- 5th BatchNorm2d layer
			- 9th BatchNorm2d layer
			- 13th BatchNorm2d layer

TABLE I: Dataset, model information, and split points used in the experiments.

predictions, so we train the models on these two types of vectors respectively. The corresponding models, their accuracy, and the selected split points are shown in Table I.

- **UrbanSound8K:** The UrbanSound8K dataset is a widely used collection of audio recordings designed for urban sound classification research. It contains 8,732 labeled sound excerpts, drawn from 10 classes of urban sounds such as car horns, and sirens. UrbanSound8K is particularly useful for tasks like environmental sound classification in real-world scenarios. We use a model consisting of 6 convolutional layers and 2 fully connected layers as the target model.
- **Google speech commands:** The Google Speech Commands dataset is a collection of audio recordings designed for training models on spoken word recognitions. It contains thousands of one-second-long utterances of short words and commands such as “yes”, and “no” spoken by a variety of speakers. It is commonly used in speech recognition research and applications like voice-controlled systems. We use a ResNet18 [17] as the target model for audio CRS.

B. Metrics and Parameters Setting

We use three metrics to evaluate the performance of InverCRS in both white-box and black-box scenarios.

- **Signal-to-noise Ratio (SNR):** SNR is a measure of the strength of a signal compared to the level of noise. A higher SNR indicates a clearer audio signal with less noise, leading to better quality, while a lower SNR means more noise relative to the reconstructed audio signal.
- **Mean Squared Error(MSE):** In this paper, we first convert the audio into MFCC/Mel-spectrogram vector and then recognize it. We used MSE to measure the difference between MFCC/Mel vectors of reconstructed audio and the original audio to evaluate the performance of InverCRS.
- **Classification Gap:** The classification gap is used to evaluate the gap in CRS classification between the reconstructed and the original audio. It is used to evaluate whether the reconstructed audio can effectively fool the CRS. A smaller classification gap indicates better performance of InverCRS.

Parameters Setting. We set the learning rate lr of attack model G is 0.001, and the number T of iterations is 100. When the split point is located in the shallow layer of the target model, we set the equilibrium coefficient λ in Equation (4) to a very small value (e.g., a positive real number close to zero), while if the split point is located in deep layers, we increase λ proportionally up to 3. The complexity of the generative attack

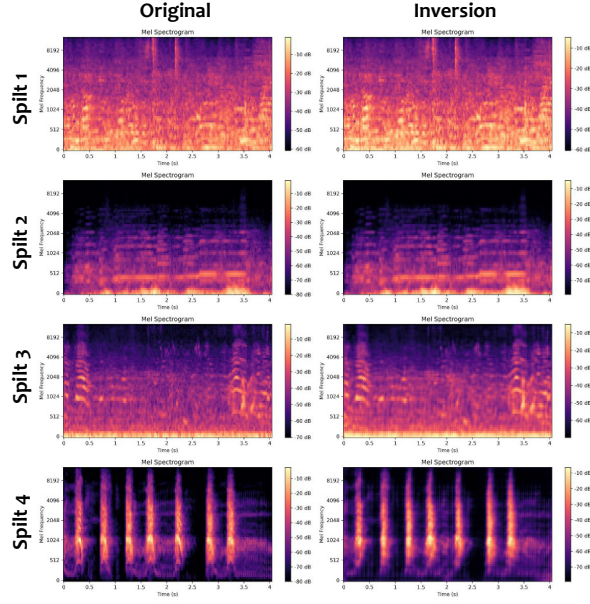


Fig. 3: At different split points, the visualization results of Mel-spectrogram of original audio and reconstructed audio for the dataset UrbanSound8K in white-box scenario.

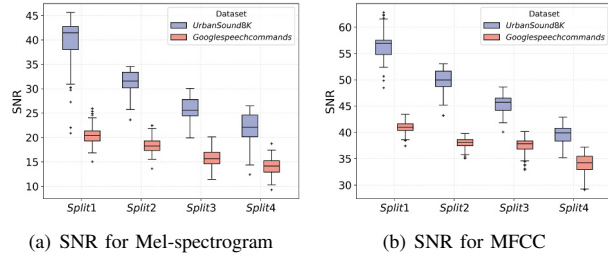


Fig. 4: The SNR numerical results of InverCRS in the white-box scenario at different split points.

model G increases as the size of the intermediate result. We conduct the experiments on a server with an Nvidia RTX 4090 GPU and an Intel(R) Xeon(R) Gold 6242 CPU @ 2.80GHz.

C. InverCRS in white-box scenario

This section evaluates the performance of InverCRS in a white-box scenario. Specifically, we will observe the effect of split points at different depths in CRS on InverCRS.

Figures 3 and 5 present the Mel-spectrogram visualization results of InverCRS attacks at various split points for the UrbanSound8K and Google Speech Commands datasets, respectively. Notably, the depth of the split points increases progressively from 1 to 4. The reconstructed audio generated by InverCRS across all four split points is visually almost indistinguishable from the original audio. However, it can be observed that the audio reconstructed at the fourth split point contains more noise compared to reconstructions at the shallower split points. Figures 4 and 6 provide the numerical results of SNR and MSE for InverCRS under white-box scenarios. These results demonstrate that as the split point moves

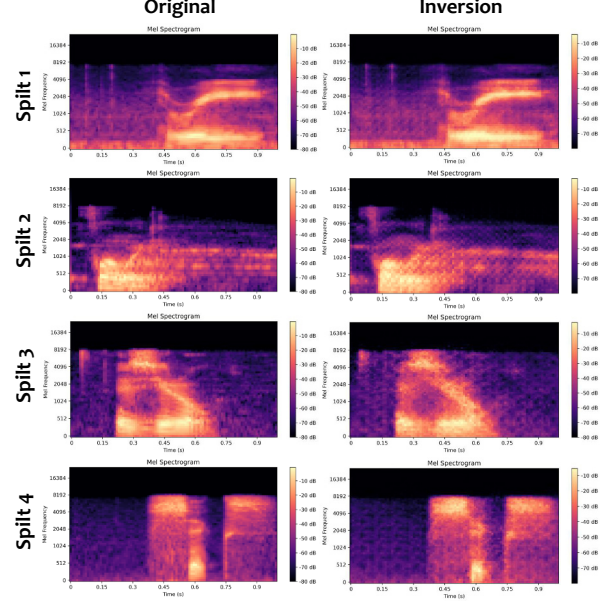


Fig. 5: At different split points, the visualization results of Mel-spectrogram of original audio and reconstructed audio for the dataset Google speech commands in white-box scenario.

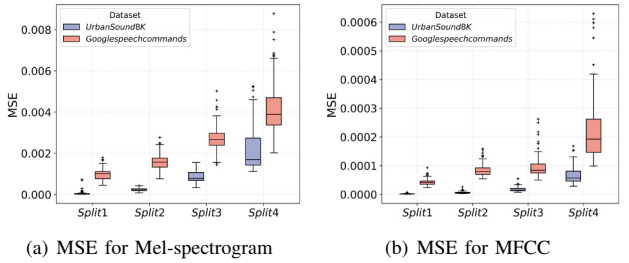


Fig. 6: The MSE numerical results of InverCRS in the white-box scenario at different split points.

deeper into the NN, the reconstructed audio exhibits increased noise. This is because deeper NN layers output intermediate results with reduced original semantic information, resulting in lower-quality reconstructed audio. Table II lists the classification gaps between the original and reconstructed audio at different split points. It is evident that InverCRS achieves better performance on the Google Speech Commands dataset compared to UrbanSound8K. This difference can be attributed to the higher classification accuracy of the ResNet18 model on the Google Speech Commands dataset. Specifically, each split point's intermediate result can be interpreted as a form of feature extraction. The superior accuracy of ResNet18 suggests that its decision boundaries are more precise and exhibit stronger generalization. Consequently, the features corresponding to the input data are positioned further from the decision boundary. When the attack model is well-trained on such features, there is a high likelihood that the reconstructed audio will remain within the correct decision space of the target model, even if the features of the reconstructed audio contain

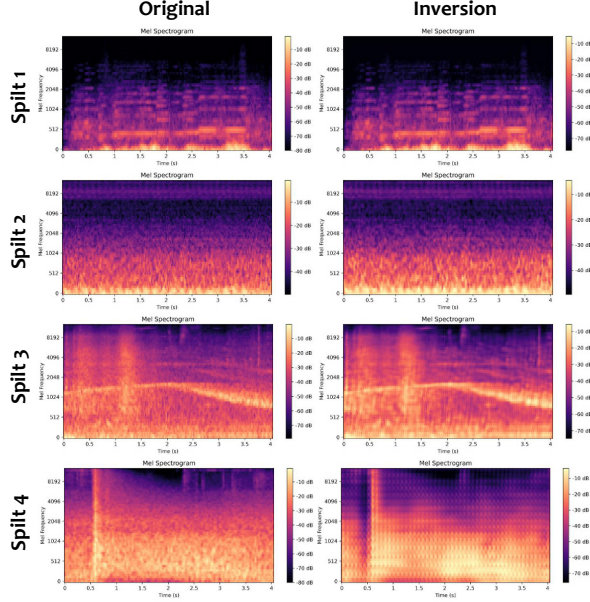


Fig. 7: At different split points, the visualization results of Mel-spectrogram of original audio and reconstructed audio for the dataset UrbanSound8K in the black-box scenario.

TABLE II: In the white-box scenario, the classification gap of original audio and reconstructed audio.

Dataset	Vector	Split 1	Split 2	Split 3	Split 4
UrbanSound8K	Mel	3.97%	8.23%	9.73%	9.93%
	MFCC	2.99%	2.99%	4.55%	11.28%
Google speech commands	Mel	1.23%	1.51%	1.51%	2.20%
	MFCC	2.11%	2.74%	3.05%	6.81%

substantial noise. Thus, the classification gap for Google Speech Commands is smaller than that for UrbanSound8K. It is also important to highlight the primary objective of CRS, which is to offload the computational overhead of audio recognition to the server. While placing the split point at a deeper layer of the NN can mitigate the threat of InverCRS, this approach significantly increases the computational overhead on the client side. When the split points are located in shallow layers, such as split points 1 and 2, the classification gap is generally within 5%. This suggests that audio reconstructed by InverCRS can effectively fool the target CRS when the split point is positioned at shallower layers.

D. InverCRS in black-box scenario

In this section, we explore the performance of InverCRS in the black-box scenario. We set the parameter m in NES to 25, that is, 25 sets of Gaussian noise are symmetrically sampled for each data during gradient approximation via NES. The search variance σ is set to 0.001.

Figures 7 and 8 depict the Mel-spectrogram visualization results of InverCRS at various split points. The results demonstrate that the shallower layer where the split point is

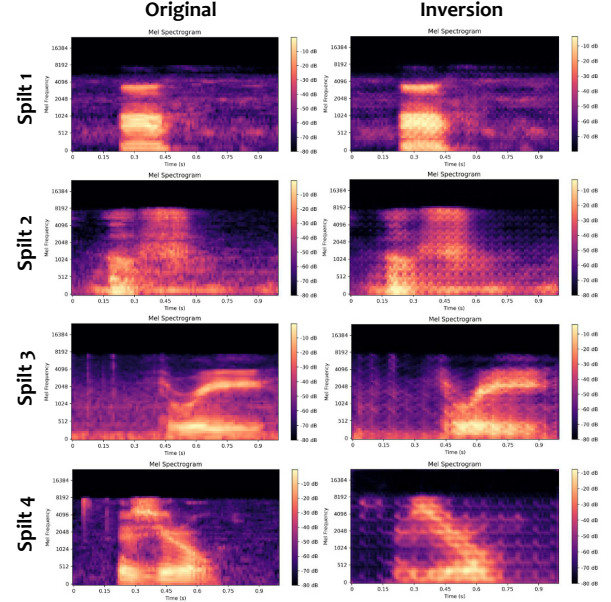


Fig. 8: At different split points, the visualization results of Mel-spectrogram of original and reconstructed audio for Google speech commands in the black-box scenario.

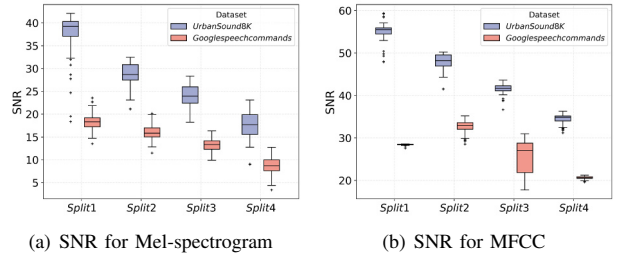


Fig. 9: The SNR results of InverCRS in the black-box scenario at different split points.

located, the higher the quality of the audio reconstructed by InverCRS. Notably, InverCRS achieves superior reconstruction quality in inversion attacks on the UrbanSound8K compared to the Google Speech Commands. Specifically, checkerboard artifacts at split points 2 and 3 are less pronounced in UrbanSound8K. However, in comparison to the white-box scenario, the audio reconstructed by InverCRS in the black-

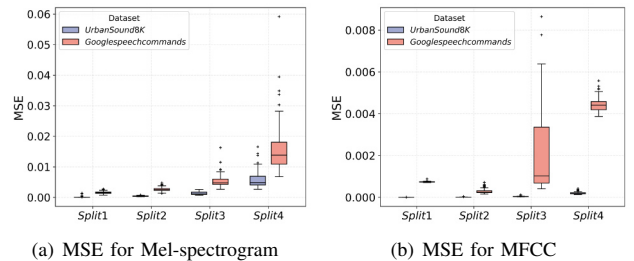
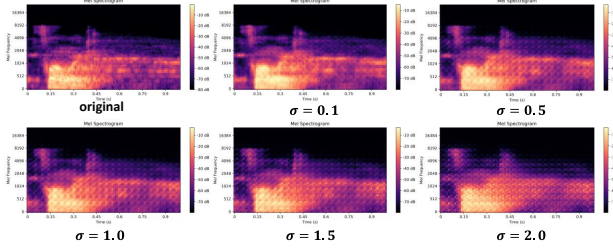


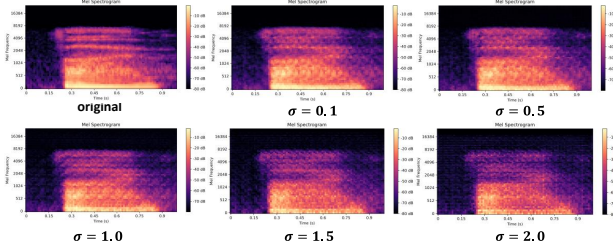
Fig. 10: The metric of InverCRS in recovering audios in the black-box scenario for different split points.

TABLE III: In the black-box scenario, the classification gap of original audio and reconstructed audio.

Dataset	Vector	Split 1	Split 2	Split 3	Split 4
UrbanSound8K	Mel	5.47%	13.93%	12.72%	17.67%
	MFCC	4.15%	5.53%	15.72%	23.09%
Google speech commands	Mel	2.87%	2.28%	3.19%	4.35%
	MFCC	3.32%	5.00%	8.01%	11.23%



(a) The result of split point 1 under differential privacy.



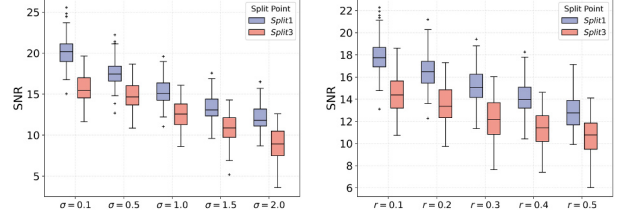
(b) The result of split point 3 under differential privacy.

Fig. 11: The Mel-spectrogram visualization results of InverCRS under various σ of differential privacy.

box scenario contains more noise. Figures 9 and 10 present the numerical results of SNR and MSE. These results align with the visual observations: the performance of InverCRS deteriorates as the split point moves deeper layer. When the split points are located at shallower layers, specifically split points 1 and 2, the MSE values are less than 0.01, indicating that InverCRS achieves excellent audio inversion performance under the black-box scenario. Table III tabulates the classification gap between the original and reconstructed audio. With the deepening of the split point, the classification gap gradually increases, and under the Google speech commands, the classification gaps are small and most of them are less than 10%. However, while placing the split point at deeper layers can effectively mitigate InverCRS, this approach introduces a substantial increase in computational overhead for the client.

E. Potential Defenses

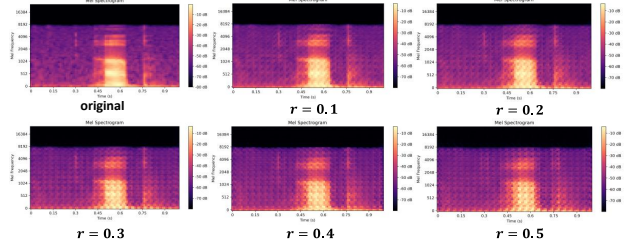
The previous experiment results have demonstrated that InverCRS can be effectively mitigated when the split point is located at deeper layer. However, this requires the client to bear a significant computational overhead, contrary to the original purpose of CRS. This section considers another two potential defenses: differential privacy and dropout.



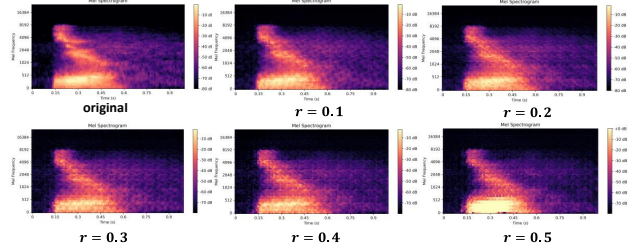
(a) SNR under differential privacy

(b) SNR under dropout

Fig. 12: The SNR results of InverCRS under two defenses.



(a) The result of split point 1 under dropout.



(b) The result of split point 3 under dropout.

Fig. 13: The Mel-spectrogram visualization results of InverCRS reconstructed audio under various r of dropout.

Differential Privacy. Differential privacy has been extensively employed to protect data privacy by adding random noise to the relevant data. We propose defending against InverCRS by introducing Gaussian noise to the intermediate results. Finally, the malicious server receives the noisy intermediate result $\tilde{I}_i = I_i + \epsilon$, rather than the exact value I_i , where ϵ follows the Gaussian noise $\epsilon \sim \mathcal{N}(0, \sigma^2)$. We set up five values for σ , $\sigma = 0.1, 0.5, 1.0, 1.5, 2.0$, where a larger value of σ indicates a higher level of noise addition.

Dropout. In addition to differential privacy, we explore the dropout as the other defense. Specifically, the client randomly sets the output of neurons at the split points to zero, thereby preventing the attacker from accessing precise intermediate results. After dropout, the intermediate result I_i become $I_i^{drop} = \text{dropout}(I_i, r)$, where r means each element of I_i is randomly assigned a value of 0 with probability r . We set five sets of r values, $r = 0.1, 0.2, 0.3, 0.4, 0.5$ to explore the impact of r on the performance of InverCRS.

We evaluate the defenses in the white-box scenario on the Mel-spectrogram vectors of the Google speech commands dataset. We select split points 1 and 3 as representative split points. Figure 11 and 13 show the visualization results of audio

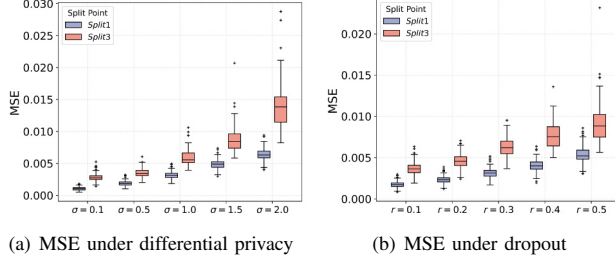


Fig. 14: The MSE results of InverCRS under two defenses.

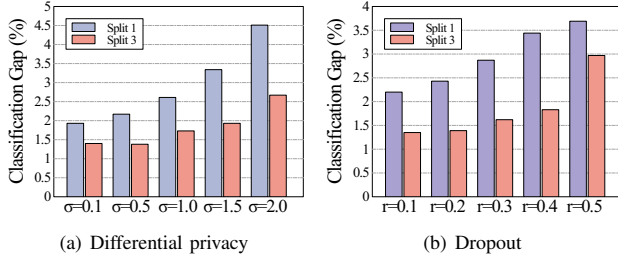


Fig. 15: The classification gap under two defenses.

reconstructed by InverCRS under different σ and r . We can conclude that as σ or r increases, the inversion audio contains more and more noise, the main reason is that more and more features contained in the intermediate results are confused by noise or abandoned. Figure 12 and 14 demonstrate the numerical results for SNR and MSE, respectively. Compared to the results without defenses presented in Figure 4 and 6, the audio reconstructed by InverCRS contains more noise under both defenses, which proves the effectiveness of the two defenses. However, combined with the visualization results, we observe that when the values of σ and r are small, InverCRS can still reconstruct high-quality audio. Figure 15 illustrates the classification gap between the reconstructed and the original audio under the two defenses. It can be observed that the classification gap is always controlled within 5% under different σ and r , which indicates that the fake audio reconstructed by InverCRS can still effectively fool CRS under these two defenses. Additionally, since differential privacy and dropout perturb the intermediate results, they will affect the accuracy of CRS, that is, the perturbed intermediate results may be misclassified. Figure 16 illustrates the impact of the two defenses on the CRS accuracy. With the increase of σ and r , the accuracy gap on normal audio between the CRS under defense and the original CRS gradually increases, even reaching 30%, in which case CRS will not be available. The defenses lead to a trade-off between privacy and accuracy. Moreover, the closer the split point is to the input layer, the larger the accuracy gaps of the two defenses. However, to alleviate the computational overhead of the client, CRS often places the split point in the first few layers, so these two defenses cannot effectively resist InverCRS due to the intolerable accuracy sacrifice of CRS.

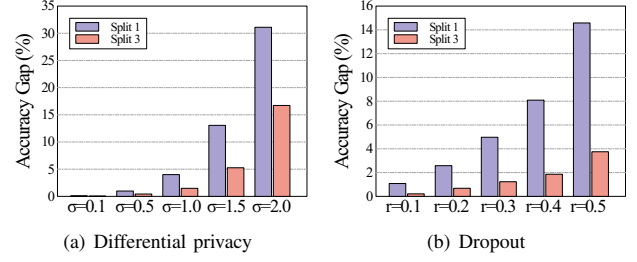


Fig. 16: The accuracy gap under two defenses.

VI. CONCLUSION

In this paper, we introduce InverCRS, the first audio inversion attack targeting CRS. This attack highlights the privacy vulnerabilities inherent in CRS systems, showing that an attacker can reconstruct the original audio from intermediate results. InverCRS is a generative inversion attack in which the attacker trains an attack model without the need for access to the target model's training dataset or any dataset with the same distribution. Once the attack model is trained, the attacker can reconstruct the original audio by simply inputting the new intermediate results instead of intensive optimization processes. Additionally, InverCRS employs a natural evolution strategy to approximate gradients, enabling its application in the black-box scenario. Extensive experiments validate the effectiveness of InverCRS, and its robustness is further evaluated through two potential defense strategies.

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